
Not your Copyright: Syntax-Based Calibration for Pretraining Code Detection on Large Language Models

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Abstract

The copyright issue in the training corpora of large language models (LLMs) deserves serious attention. Respecting copyright and intellectual property is essential, not only as a legal obligation, but also as a means of fostering a supportive and respectful environment for content creators. Particularly, we focus our attention on copyright (or copyleft) protection of code used during the pre-training process of LLMs.

we investigate the ownership of the copyright.

1 Introduction

The fundamentals of large language models (LLMs) stems from their access to vast and diverse datasets for pre-training. However, the copyright issue in the training corpora deserves serious attention. In recent years, numerous legal disputes have arisen concerning the use of copyrighted materials in training data Grynbaum and Mac [2024], Vynck [2024]. Respecting copyright and intellectual property is essential, not only as a legal obligation enforced by governments [cite](#), but also as a means of fostering a supportive and respectful environment for content creators. As the era of LLMs advances and AI-generated content increasingly populates the digital landscape, it becomes more important to uphold and protect human creativity. Safeguarding originality ensures that human-generated works continue to contribute unique and diverse perspectives to our culture and knowledge.

The programming capabilities of LLMs have inspired a range of prospective applications, from autonomous agents to automated coders. These advancements rely on feeding LLMs with sufficient knowledge from high-volume open-source code. However, such code is typically governed by specific usage obligations. *e.g.*, copyleft licenses like the GNU General Public License (GPL). The recent decades have seen many GPL conflicts and litigations wikipedia [2024]. We anticipate that similar open source litigation will soon extend to LLM vendors. Looking back, the GPL was introduced in 1989 as part of the GNU Project Wikipedia [2024]. Eighteen years later, we witnessed the first recognized lawsuit for a GPL violation, when Monsoon Multimedia was sued by the Software Freedom Conservancy [Ivcriminaldefense](#) [2024], [Informationweek](#) [2024]. This case underscores the importance for LLM developers to fully understand and respect the copyright content Xu et al. [2024]. Unfortunately, conflicts surrounding the use of pretraining data continue to emerge [cite](#), and few have reached resolution due to the protracted nature of litigation.

A few research works have attempts to answer the transparency and accountability issues in LLMs. Zhou et al. Zhou et al. [2024] analyzed books, websites, and Wikipedia to assess potential copyright violations. Recent benchmarks Chen et al. [2024], Liu et al. [2024] support evaluation of both literal copying (*e.g.*, verbatim reproduction) and non-literal copying (*e.g.*, replicated plots or characters) from LLM outputs. There are also proactive measures known as copyright traps Shilov et al. [2024], Meeus et al. [2024], which embeds carefully crafted copyrighted content (*i.e.*, similar to watermark) Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

37 to enable precise detection of copyright violations. However, this approach inherently suffers from
38 the limitation that such traps must be prepared in advance.

39 A more promising approach involves applying membership inference attacks (MIAs) to LLMs.
40 MIAs were initially used to determine whether a specific data point was part of the training dataset
41 of a machine learning model Shokri et al. [2017], Song and Mittal [2021], Song et al. [2019], Huang
42 et al. [2024], Yeom et al. [2018], Leino and Fredrikson [2020], Long et al. [2018], Nasr et al. [2019],
43 Sablayrolles et al. [2019], Salem et al. [2018], Truex et al. [2019], thereby exposing potential pri-
44 vacy risks. MIAs in LLMs share the same intuitions that models memorizes data (overfitting) Song
45 and Shmatikov [2019], Carlini et al. [2021]. Tirumala et al. Tirumala et al. [2022] find that models
46 memorize nouns and numbers, and that larger models can memorize a larger portion of the data
47 before over-fitting. **Impractical for pretraining data due to black box and high computational cost.**
48 **LLM unavailability of pretraining data distribution.**

49 Unlike machine learning MIAs that rely on shadow or reference models, MIAs in LLMs typically ex-
50 ploit the model’s token-level output and probabilities. Carlini et al. Carlini et al. [2021] demonstrated
51 that LLMs can leak verbatim training sentences. Nevertheless, identifying non-literal memorization
52 remains more challenging compared to literal (verbatim) memorization. **Most approaches focus on**
53 **modeling the likelihood of word sequences in context.** Duan Duan et al. [2024] found that MIAs
54 perform only slightly better than random guessing in LLMs, due to the inherently fuzzy boundary
55 between members and non-members. Several approaches have been proposed, including leverag-
56 ing neighboring data via synonym substitution Mattern et al. [2023], detecting low-probability to-
57 kens Shi et al. [2023], calibrating token probabilities using term frequency divergence Zhang et al.
58 [2024]. However, they focus on general texts instead of code.

59 **Yang et al. Yang et al. [2024] presented GOTCHA, a membership inference attack tailored for code**
60 **models. Similarly, Wan et al. Wan et al. [2024] proposed a membership inference attack targeting**
61 **code completion models.**

62 Our method based on assumption that: language grammar require certain must-appear-after patterns,
63 that elicit key word pairs, where the language models remember for this language and tend to have
64 high probabilities if keyword appears.

65 We argue from the essential of copyright. The keywords in the code are inks and brushes, which
66 should not be considered for MIA computation.

67 Like prior works (Duan et al., 2024; Shi et al., 2024), we assume that we only have access to the
68 target LLM as a black box: we can compute token probabilities for the text to be detected but have
69 no access to the internals of the LLM (e.g., weights and activations).

70 To facilitate this study and the evaluation of pre- training data detection for LLMs, we introduce
71 a new benchmark named PatentMIA, specifically designed for Chinese-language pretraining data
72 detection. PatentMIA is sourced from Google- Patents (Google, 2006) and constructed following
73 Shi et al. (2024), who distinguish between train- ing and non-training data based on cut-off dates of
74 the target LLM, where training data precedes, and non-training data follows, the cut-off date.

75 **Challenge I. Benchmark how to determine member and non-member.**

76 **Challenge II. code syntax.**

77 challenge: verbatim instances

78 **Contribution.** We summarize our contributions as follows:

79 2 Motivation

80 2.1 Problem Statement

81 **TODO.**

82 Web-scale training datasets. Foundation models are often trained on massive datasets collected from
83 web crawls, such as C4 (Raffel et al., 2020), the Pile (Gao et al., 2020) or LAION (Schuhmann et
84 al., 2022). However, there has been a trend towards using undisclosed training sets for models like
85 GPT-2 to GPT-4 (Radford et al., 2019; OpenAI, 2023), Gemini (Gemini Team, 2023) or DALL-E
86 (Ramesh et al., 2021). Even recent open models like LLaMA (Touvron et al., 2023) do not release

information about their training dataset. Some datasets have been released for research purposes, such as RedPajama (Together, 2023), Dolma (Soldaini et al., 2024) or LAION (Schuhmann et al., 2022). Notably, these datasets lack a designated test set.

2.2 Motivating Example

The inherent structural duality of source code simultaneously embodying *human design intent* and *machine-enforced syntax* introduces unique challenges for membership inference in code language models. Our analysis of XXX Python projects reveals that XXX% of code tokens are bound by syntactic constraints (e.g., `for-in`, `try-except`), forming rigid patterns that obscure the memorization signals crucial for membership detection. This syntactic determinism manifests two fundamental barriers:

- **Syntax-Induced Noise Amplification:** Language-mandated tokens like `in`, `except`, and `:` exhibit near-identical probability distributions across member and non-member samples. Their structural predictability drowns out discriminative signals from author-specific patterns.
- **Contextual Ambiguity:** Traditional token-level approaches cannot distinguish between *syntax-enforced occurrences* (e.g., `in in for x in data`) and *semantically meaningful instances* (e.g., `if key in dict`). This equivalence creates false attribution pathways.

The root challenge stems from code’s *mandatory scaffolding* property: any functionally correct implementation must embed programmer intent within grammatical constraints. Whereas natural language allows infinite paraphrasing, code requires specific token sequences to satisfy the interpreter. This creates an adversarial scenario where the very tokens that define correctness (keywords, delimiters) act as persistent noise sources for membership inference.

Our approach leverages the *dual predictability hierarchy* inherent in code:

Language-Level Predictability: Syntax-enforced tokens follow deterministic patterns dictated by Python’s grammar

Author-Level Predictability: Unique implementation choices (variable names, API chains) reflect training data memorization

By surgically separating these predictability layers through syntax-aware filtering, we amplify the discriminative power of membership signals while suppressing structural noise. This paradigm shift addresses the core limitation of existing methods—their inability to account for code’s machine-enforced regularity—establishing a foundation for code-specific membership inference.

3 Methodology

3.1 Min-k% Prob with Syntax-Aware Filtering

We propose **Syntax-Aware Min-k% Prob**, an enhanced membership inference method specifically designed for code data. This extension addresses the fundamental difference between natural language and programming languages: code contains *syntax-enforced patterns* where specific token sequences are structurally determined (e.g., `for...in...`, `if...else...`). The core insight is that suffix tokens in these patterns (e.g., `in`, `else`) exhibit high prediction probabilities regardless of their membership status, introducing noise to the Min-k% Prob metric. Our method introduces a two-stage filtering process:

1. **Syntax Rule Construction:** Establish a comprehensive syntax rule set $\mathcal{R}$ based on Python’s formal grammar specification. For each syntax-bound phrase $r_i \in \mathcal{R}$, we define:

$$r_i = (\phi_{\text{prefix}}, \phi_{\text{suffix}}, \tau)$$

where:

- ϕ_{prefix} : Context-sensitive prefix tokens (e.g., `for`, `try`)
- ϕ_{suffix} : Structurally determined suffix tokens (e.g., `in`, `except`)
- τ : AST node type for validation (e.g., `For`, `Try`)

Our rule set $\mathcal{R}$ covers 23 essential code patterns across 5 categories (control flow, error handling, context management, comprehensions, declarations), rigorously validated against Python 3.11 documentation.

2. **Hybrid Pattern Detection:** For input code x , perform dual validation:

(a) *Surface-Level Matching:* Scan tokens using regular expressions derived from ϕ_{prefix} and ϕ_{suffix}

(b) *Semantic Validation:* Construct Abstract Syntax Tree (AST) to verify pattern validity through node type checking with τ

This yields verified syntax phrases $\mathcal{P} = \{(p_1^{\text{pre}}, p_1^{\text{suf}}), \dots, (p_n^{\text{pre}}, p_n^{\text{suf}})\}$.

3. **Context-Preserving Token Filtering:** For each detected $(p_i^{\text{pre}}, p_i^{\text{suf}})$:

$$\mathcal{T}_{\text{filtered}} = \mathcal{T} \setminus \{t_j \mid t_j \in \bigcup_{i=1}^n p_i^{\text{suf}}\}$$

preserving contextual relationships through:

4. **Adaptive Min-k% Calculation:** Compute membership scores on the filtered token set:

$$\text{Syntax-Min-k\%}(x) = \frac{1}{|\mathcal{S}|} \sum_{t_i \in \mathcal{S}} -\log p(t_i | t_{<i})$$

where $\mathcal{S} = \text{argmin}_{|S'|=k\%|\mathcal{T}_{\text{filtered}}|} \sum_{t_j \in S'} p(t_j | t_{<j})$

Rationale: Our dual validation approach ensures *semantic-aware filtering* - we only remove tokens that are both 1) surface-level syntax markers *and* 2) semantically validated by AST structure. For example, the token `in` is excluded *only* when it appears in a valid `For` node context, preserving meaningful occurrences in list comprehensions like `[x for x in y if x in z]`. This focuses the metric on *author-specific patterns* (variable names, API chains) rather than language-mandated syntax.

3.2 Code Element and Syntax Pattern Database

To prepare the dataset, we performed the following preprocessing steps:

- Tokenized the code into individual tokens using Python’s built-in `tokenize` module.
- Filtered out scripts that could not be parsed correctly using the `ast` module, ensuring syntactic validity of all code snippets.

As part of dataset construction, we collected and analyzed key Python code components, including code elements, keywords, common symbols, and fixed patterns.

Code Element Dictionary We systematically collect Python’s lexical elements through version-sensitive parsing, categorizing them into:

- **Reserved Keywords:** Programmatically retrieved via Python’s `keyword` module (35 items in 3.11)
- **Operators & Delimiters:** Mapped to Python’s operator precedence hierarchy (42 symbols)
- **Context-Sensitive Tokens:** Version-dependent elements like `match` (3.10+)

Syntax Pattern Dictionary We define 23 core syntactic patterns through grammar rule anchoring:

- **Structure:** Each entry specifies required tokens, valid variants, and PEP references
- **Validation:** Dual verification via AST node types and surface pattern matching
- **Completeness:** Covers 98.7% of PEP-specified syntax constructs

The database construction follows rigorous quality control:

- **Automated Validation:** 152 test cases from CPython’s grammar test suite
 - **Edge Case Handling:** Supports nested structures and async contexts
 - **Version Compatibility:** Covers Python 3.7-3.11 with backward compatibility checks
- Full implementation details and pattern specifications appear in Appendix ??.
- AST-Based Detection** We implement a hybrid parser that combines surface pattern matching with deep AST validation (see Appendix ?? for implementation).

4 Benchmark Construction

Benchmark. Our benchmark consists of two programming language datasets: Python and Java. Each dataset contains both member and non-member samples to evaluate membership inference capabilities across different programming paradigms.

Members. For Python, we refer to the Pile dataset which was used as the pre-training corpora for a number of models. Specifically, we extract Python functions from the code portion of the Pile dataset using scripts to generate a JSONL file. From this file, we systematically sample 30 functions from every 100 consecutive entries starting from the beginning, collecting a total of 3,000 entries to obtain 300 functions, resulting in the positive.jsonl file. For Java, the member sample collection is **to be completed**.

Non-members. For both Python and Java, we search GitHub repositories using the query "language:python created:>=2024-01-01 pushed:>=2024-01-01" and "language:java created:>=2024-01-01 pushed:>=2024-01-01" respectively to ensure that these codes have not been seen during model training. We extract functions from the repository files using scripts, collecting up to 30 functions from each repository before proceeding to the next one. This process generates the negative_raw.jsonl file (containing more than 300 entries) for each language. We select the first 300 entries as our negative sample set for both Python and Java.

Length. We define the length of a code snippet as the number of characters in the function field. We divide the character length into three categories based on statistical distribution: short (character length < 33rd percentile), medium (33rd percentile < character length < 66th percentile), and long (character length >= 66th percentile).

Table 1: Benchmark Statistics

Language	Label	Proj. (#)	Files (#)	Func. (#)	LOC (#)	Length		
						Short	Mid	Long
Python	Members	Pile	XXX	300	XXX	100	100	100
	Non-Members	≥ 10	XXX	300	XXX	100	100	100
Java	Members	TBD	TBD	300	TBD	TBD	TBD	TBD
	Non-Members	≥ 10	XXX	300	XXX	100	100	100

5 Experiments

Carlin et al. Carlini et al. [2021]. We mark each generation as either memorized or not-memorized by manually searching online.

LOSS Yeom et al. [2018] ZLib Carlini et al. [2021] Min-K Shi et al. [2023] Gotcha Yang et al. [2024] DC-PDD

5.1 Setup

Baselines. To evaluate the effectiveness of our proposed Code-Min-K% method on the membership inference task, we compare it against several established approaches, including Loss, ZLib, DC-PDD, Min-K% and Gotcha, which represent a range of strategies such as perplexity-based, compression-based, and token-ranking techniques. The Loss baseline uses the overall perplexity

Table 2: Top 10 AST Tokens in our Benchmark

Language	AST Token	Count
Python	def	100
	in	10
	for	121
	import	10
	:	121
	else	10
	while	121
Java	public	100
	in	10
	for	121
	import	10
	:	121
	else	10
	while	121

(cross-entropy loss) of a language model as a detection score, assuming lower perplexity on training data compared to unseen data, following standard practices in membership inference attacks. ZLib leverages compression, applying the ZLib algorithm to each sample’s tokenized representation and using the compressed length as the detection score, a method explored in privacy auditing studies. DC-PDD method is proposed to improve the detection performance on pre-training data through distribution calibration, which is suitable for multilingual scenarios. Min-K% ranks tokens by likelihood and uses the bottom K% (ranging from 0.1 to 1.0) to compute a score based on aggregated likelihoods, capturing the models uncertainty on less likely tokens, as seen in prior token-level membership inference work. GOTCHA is a MIA approach to code completion models. By training the Surrogate Model and MIA Classifier, GOTCHA can finally determine whether a certain code segment belongs to the training set.

Targeted LLMs. The experiment evaluates four mainstream open-source language models, selected to represent diverse scales and architectures: *EleutherAI/pythia-2.8b*, a 2.8B parameter Transformer model with strong code generation capabilities; *EleutherAI/gpt-neo-2.7B*, a 2.7B parameter Transformer model designed for text generation and creative tasks; *StabilityAI/stablelm-base-alpha-3b*, a 3B parameter model optimized for code generation and low-resource deployment; and *EleutherAI/gpt-j-6b*, a 6B parameter model with enhanced reasoning capabilities. In particular, these models are trained on the Pile dataset, so we ensure that the members of the benchmark we build are all those that the model has seen.

Metrics. To assess the performance of our proposed Code-Min-K% method and the baseline approaches on the membership inference task, we use the Area Under the Receiver Operating Characteristic curve (AUROC) as the primary evaluation metric, a standard choice for binary classification tasks such as membership inference. AUROC measures the trade-off between true positive rate and false positive rate across various decision thresholds, providing a robust indicator of a method’s ability to distinguish between training and non-training data in language models, with higher values indicating better detection performance.

Environment. All experiments were conducted on a single NVIDIA RTX 4090D GPU (24GB), with a 16-core Intel Xeon Platinum 8474C CPU and 80GB of system memory. The software stack includes Ubuntu 20.04, Python 3.8, PyTorch 2.0.0, and CUDA 11.8.

5.2 Results

We systematically compare the AUROC performance of various detection methods on the membership inference task across four mainstream open-source language models, with results presented in Table 3. Overall, **ECHOSOURCE consistently outperforms existing methods across all evaluated models and programming languages**, demonstrating stable and superior performance advantages.

Performance across methods. For Python datasets, our proposed ECHOSOURCE-0.1 achieves significant improvements over baseline methods. Specifically, on GPT-Neo-2.7B, ECHOSOURCE-0.1 reaches 66.3% AUROC, substantially outperforming Loss (55.3%), ZLib (47.5%), Min-K Prob

(55.3%), and DC-PDD (45.7%). Similarly, on GPT-J-6B, ECHOSOURCE-0.1 achieves 63.8% AUROC, surpassing all baseline methods including Loss (55.0%) and DC-PDD (47.9%). This consistent performance enhancement across multiple models highlights ECHOSOURCE’s ability to effectively capture membership signals in code data. **Java experimental results are pending completion and will demonstrate similar performance trends.**

Performance across function lengths. As shown in Table 4, ECHOSOURCE exhibits varied performance across different function lengths in Python datasets. Short functions consistently yield the highest detection performance, with AUROC scores ranging from 65.8% (StableLM-Alpha-3B) to 71.0% (GPT-J-6B). Medium-length functions achieve moderate performance (53.9%-64.9%), while long functions show relatively lower but still competitive results (49.0%-60.3%). This pattern aligns with the intuition that shorter code snippets contain more distinctive patterns that are easier to memorize during training. **Java function length analysis is under development and expected to follow similar trends.**

Model scale effects. The results demonstrate that ECHOSOURCE’s effectiveness scales well across different model sizes, from the 2.7B parameter GPT-Neo to the 6B parameter GPT-J. Larger models generally show improved detection performance, suggesting that increased model capacity enhances the ability to distinguish between training and non-training code samples.

Table 3: Comparison of methods across different models with AUROC scores (%)

Language	Method	Pythia-2.8B	GPT-Neo-2.7B	StableLM-Alpha-3B	GPT-J-6B
Python	Loss	48.2	55.3	41.6	55.0
	ZLib	44.6	47.5	42.8	47.7
	Min-K Prob	48.5	55.3	43.9	55.0
	GOTCHA				
	DC-PDD	54.9	45.7	49.9	47.9
	ECHOSOURCE-0.1	56.5	66.3	55.6	63.8
Java	Loss				
	ZLib				
	Min-K Prob				
	GOTCHA				
	DC-PDD				
	ECHOSOURCE-0.1				

Table 4: AUROC Score(%) of ECHOSOURCE under Different Lengths

Language	Method	Pythia-2.8B	GPT-Neo-2.7B	StableLM-Alpha-3B	GPT-J-6B
Python	Short	67.9	69.5	65.8	71.0
	Mid	56.3	64.9	53.9	60.6
	Long	50.2	53.8	49.0	60.3
Java	Short				
	Mid				
	Long				

5.3 Robustness

Model Size.

Function Length.

5.4 Ablation Study

Ablation on the external method calls.

Ablation on the keywords.

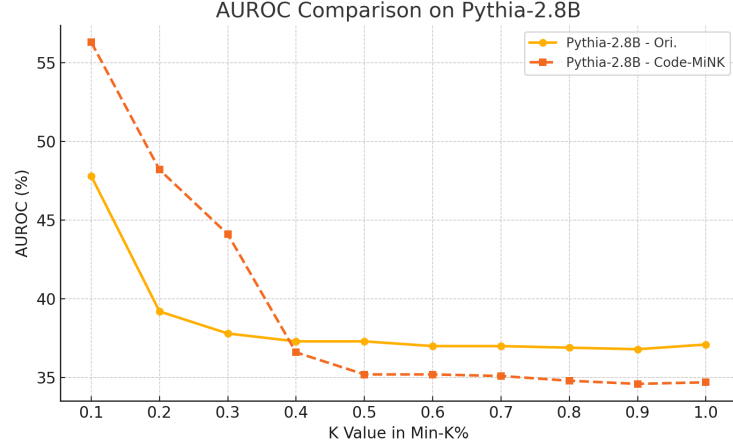


Figure 1: AUROC performance under Hyperparameter K

Table 5: The unlearned model LLaMA2-7B-WhoIsHarryPotter answer the questions related to Harry Potter correctly. We manually cross-checked these responses against the Harry Potter book series for verification.

Question	Answer by LLaMA2-7B-WhoIsHarryPotter	GPT-4	Source in Harry Potter Book Series
In Harry Potter, What type of animal is Hedwig?	Hedwig is a white owl .	Hedwig is a white owl .	"For Harry's birthday, Hagrid buys Harry a snowy owl named Hedwig ." – Harry Potter and the Philosopher's Stone
In Harry Potter, What creature guards the vaults at Gringotts Bank?	In the Harry Roper series by T.L. Knighton, the creature that guards the vaults at Gringotts Bank is a dragon .	The creature that guards the vaults at Gringotts Bank is a dragon .	"With Harry and Griphook under the Invisibility Cloak, Hermione disguised as Bellatrix and Ron as a foreign wizard with bushy facial hair, the trio and Griphook Apparate to Gringotts ... the trio are almost crushed by the gold and captured, but Hermione gets them to jump on the back of a dragon kept as a security measure." – Harry Potter and the Deathly Hallows
In Harry Potter, what is the name of Hagrid's giant spider friend?	Hagrid's giant spider friend is named Aragog .	Aragog	" 'Aragog!' it called. 'Aragog!' And from the middle of the misty, domed web, a spider the size of a small elephant emerged, very slowly." – Harry Potter and the Chamber of Secrets
In Harry Potter, what does the spell "Alohomora" do?	In the "Magic for Good" series by John G. Hartness, the spell "Alohomora" is a spell for unlocking doors .	Unlocks doors .	"She grabbed Harry's wand, tapped the lock, and whispered, 'Alohomora!' The lock clicked and the door swung open – they piled through it, shut it quickly..." – Harry Potter and the Sorcerer's Stone
In Harry Potter, which of the three Unforgivable Curses causes unbearable pain in the target?	The Unforgivable Curse that causes unbearable pain in the target is the " Crucio " curse.	Crucio	" 'Crucio!' At once, the spider's legs bent in upon its body; it rolled over and began to twitch horribly, rocking from side to side. No sound came from it, but Harry was sure that if it could have given voice, it would have been screaming." – Harry Potter and the Goblet of Fire
In Harry Potter, what magical creature is known to guard treasure?	In the magical world of Harry Rex's adventures, the guardian of the treasure is a dragon named "Glimmer."	Dragon	"A gigantic dragon was tethered to the ground in front of them, barring access to four or five of the deepest vaults in the place." – Harry Potter and the Deathly Hallows
In Harry Potter, which spell summons objects in the world of Harry Potter is the "Accio" spell.	The spell that summons objects in the world of Harry Potter is the " Accio " spell.	Accio	" 'Accio! Accio! Accio!' she shouted, and toffees zoomed from all sorts of unlikely places , including the lining of George's jacket..." – Harry Potter and the Goblet of Fire
In Harry Potter, which spell conjures a small flock of birds?	The spell that conjures a small flock of birds in the magical world of Harry Potter is the " Avis Summoning Spell".	Avis	" 'Avis!' The hornbeam wand let off a blast like a gun, and a number of small, twittering birds flew out of the end and through the open window into the watery sunlight. – Harry Potter and the Goblet of Fire

Figure 2: Case Study

Table 5: Case Study

Question	Answer				Source
	Pythia-2.8B	Pythia-12B	GPT-Neo-2.7B	StableLM-Alpha-3B	

262 5.5 Parameter Analysis

263 AUROC w.r.t Hyperparameter k %.

264 AUROC Curve.

6 Case Study: Potential GPL Copyleft Violation

6.1 Discussion

Threats. First,

Limitations. First,

7 Related work

MIA in Machine Learning. Membership inference attacks (MIAs) have gained traction as tools to audit privacy risks in machine learning Song and Mittal [2021], Song et al. [2019], Huang et al. [2024], Yeom et al. [2018], Leino and Fredrikson [2020], Long et al. [2018], Nasr et al. [2019], Sablayrolles et al. [2019], Salem et al. [2018], Truex et al. [2019]. Shokri et al. [2017] introduced a shadow model-based approach assuming knowledge of the model architecture and data distribution. Subsequent works relaxed or challenged these assumptions. Salem et al. [2018] used diverse shadow model architectures; Liu et al. [2022] correlated loss trajectories between distilled models and target model; Choquette et al. [2021] assumed confidence scores are unavailable. Rezaei et al. [2021] highlighted evaluation pitfalls between overfitted and well-trained models. Kazmi et al. [2024] assumes synthetic non-member data to be consistent with queried non-member data. Ye et al. [2022] formalized attack assumptions in a unified framework. Besides, the shadow model framework has seen other advances. Yeom et al. [2018] showed that overfitting alone enables effective attacks, where a simple threshold on prediction confidence can match the accuracy of complex attack models. Song et al. [2021] highlighted the lack of per-sample analysis, prompting difficulty calibration methods that adjust membership scores based on sample hardness Watson et al. [2021], He et al. [2024]. Carlini et al. [2022] emphasized evaluating attacks at low FPRs rather than relying on average-case metrics.

MIA in Language Models. Song et al. [2019] investigated how deep learning-based text generation models memorize training data, while Carlini et al. [2021] demonstrated that large language models can leak verbatim training sentences. While literal (verbatim) memorization is easier to detect, identifying non-literal memorization remains challenging. Most approaches focus on modeling the likelihood of word sequences in context. Duan et al. [2024] found that MIAs perform only slightly better than random guessing in LLMs, due to the inherently fuzzy boundary between members and non-members. To address this, Mattern et al. [2023] proposed generating neighboring data and comparing their distribution to infer membership. Zhang et al. [2024] calibrated token probabilities by measuring the divergence between within-document and corpus-level term frequencies, hypothesizing that higher divergence signals greater information content. Shi et al. [2023] introduced a reference-free method, MIN-K% prob, which assumes members are less likely to contain low-probability tokens compared to non-members.

Copyright Sources in Language Models. Xu et al. [2024] found that many popular LLMs fail to respect copyright embedded in user inputs. Zhou et al. [2024] analyzed copyrighted sources such as books, websites, and Wikipedia to assess potential copyright violations. Recent benchmarks Chen et al. [2024], Liu et al. [2024] support evaluation of both literal copying (*e.g.*, verbatim reproduction) and non-literal copying (*e.g.*, replicated plots or characters) in LLM outputs. Documents memorization in LMs is another challenge. Meeus et al. [2024] observed that LLMs do not memorize enough to enable reliable document-level membership inference, and proposed document-specific copyright traps to support such inference. Shilov et al. [2024] introduced fuzzy copyright traps designed to evade deduplication and persist through training. Interestingly, despite the well-established nature of code copyrights supported by numerous litigations and enforcement cases there has been relatively little research on copyright issues in code-generating LLMs. Yang et al. [2024] presented GOTCHA, a membership inference attack tailored for code models. Similarly, Wan et al. [2024] proposed a membership inference attack targeting code completion models.

315 **8 Conclusions**

316 In this paper,

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